

Report
Of
STOCK PRICE PREDICTOR

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: **25SCS1003001650**

Name of Student: **PUSHPENDRA SINGH**

Batch: **2025-29**

2026-27

CANDIDATE'S DECLARATION

I, Pushpendra Singh, do hereby declare that the internship report titled “Stock Price Predictor” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All references have been appropriately acknowledged. I affirm that this report represents my academic and project work and has not been submitted to any other institute for any degree or diploma requirement. I further understand that I shall be solely responsible for any kind of copyright violation in this regard.

Signature: **PUSHPENDRA SINGH**

Student Name: Pushpendra Singh

Roll No.: 25SCS1003001650

ACKNOWLEDGEMENT

I am using this opportunity to express my sincere gratitude to everyone who supported and guided me throughout my internship program and project work.

I am thankful to Codec Technologies Pvt. Ltd. for providing me with the opportunity to undertake an internship as an Artificial Intelligence Intern and gain practical exposure to Artificial Intelligence, Machine Learning, data analysis, and predictive modelling.

I would also like to express my sincere gratitude to the faculty members and mentors who provided their valuable guidance, suggestions, and encouragement during the development of the Stock Price Predictor project.

The internship provided me with an opportunity to connect theoretical knowledge with practical implementation. Working on the project helped me improve my programming, analytical thinking, problem-solving, data-handling, and technical skills.

I am also grateful to my parents for their continuous support, encouragement, and blessings throughout this learning experience.

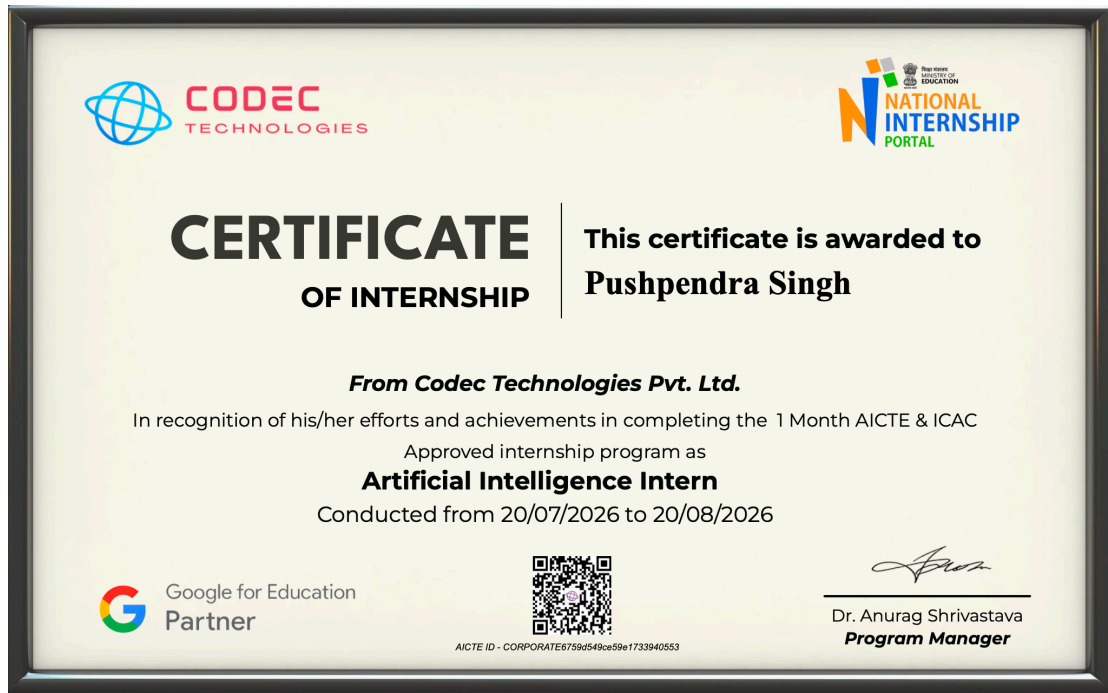
Signature: **PUSHPENDRA SINGH**

Student Name: Pushpendra Singh

Roll No.: 25SCS1003001650

INTERNSHIP COMPLETION CERTIFICATE

The supplied certificate confirms completion of the one-month AICTE & ICAC approved internship program at Codec Technologies Pvt. Ltd. as an Artificial Intelligence Intern from 20/07/2026 to 20/08/2026.



4. PROJECT DESCRIPTION

4.1 INTRODUCTION

The Stock Price Predictor project focuses on the development of a Machine Learning-based system that predicts stock prices using historical stock market data. Stock markets generate large amounts of data every trading day, including opening price, closing price, highest price, lowest price, adjusted closing price, and trading volume.

Analysing historical stock data can help identify trends and patterns in previous market movements. However, stock prices are influenced by numerous factors and can change significantly over time. This makes stock price prediction a challenging data-driven problem.

The primary objective of this project is to understand how historical stock market data can be collected, processed, analysed, and used to develop a predictive Machine Learning model.

The project follows a systematic workflow beginning with data collection and preprocessing. The historical data is then analysed to understand trends and relationships between different variables. Relevant features are selected and used to train a suitable prediction model.

After training, the model is tested using appropriate historical data and its predictions are compared with actual stock prices. This helps in evaluating the effectiveness of the developed model.

The project provides practical exposure to Python programming, data analysis, data visualization, Machine Learning, and predictive modelling. It also demonstrates how theoretical concepts can be applied to a real-world financial data problem.

Since stock prices are affected by many unpredictable factors, model predictions are estimates based on the historical information supplied to the model and should not be treated as guaranteed future prices or financial advice.

4.2 ORGANIZATION PROFILE

Codec Technologies Pvt. Ltd. is the organization where the internship was undertaken. The internship was conducted in the field of Artificial Intelligence, with the official designation of Artificial Intelligence Intern.

The offer letter specifies a Pan-India, Hybrid internship from 20 July 2026 to 20 August 2026. The program provided project and training tasks intended to provide industry-level learning and practical skills.

Organization Details:

Organization Name: Codec Technologies Pvt. Ltd.

Internship Role: Artificial Intelligence Intern

Internship Duration: 20 July 2026 – 20 August 2026

Mode: Hybrid

Location: Pan India

Field: Artificial Intelligence

Project Title: Stock Price Predictor

Program Manager: Dr. Anurag Shrivastava

The internship experience provided an opportunity to connect academic learning with practical technical work and develop a stronger foundation in Artificial Intelligence and Machine Learning.

4.3 PROBLEM STATEMENT

Stock market prices change continuously and are affected by various factors, making accurate prediction a challenging task. Historical market data contains information about previous price movements, but manually analysing large datasets can be difficult and time-consuming.

The main problem addressed by the Stock Price Predictor project is to develop a Machine Learning model that can use historical stock market data to generate predictions of stock prices.

The project requires historical data to be collected, cleaned, analysed, and transformed into a suitable format for Machine Learning. Relevant features must then be identified and an appropriate predictive model must be trained.

Another challenge is that stock market data is time-dependent. The chronological relationship between observations needs to be considered when preparing the dataset for training and testing.

Key Problems Identified:

1. Collecting suitable historical stock market data.
2. Understanding the structure of financial datasets.
3. Cleaning and preprocessing historical data.
4. Identifying relevant variables for prediction.
5. Analysing historical stock price trends.
6. Selecting an appropriate Machine Learning approach.
7. Training a predictive model.
8. Testing the model on suitable data.
9. Evaluating predicted values against actual values.
10. Identifying limitations and areas for improvement.

4.4 PROJECT OBJECTIVES

The main objective of the Stock Price Predictor project is to develop a model that predicts stock prices based on historical data.

1. To understand historical stock market data and its important variables.

2. To collect suitable historical stock price data.
3. To clean and preprocess the collected dataset.
4. To analyse historical stock price trends and patterns.
5. To identify relevant features for prediction.
6. To understand the application of Machine Learning to financial data.
7. To develop and train a suitable predictive model.
8. To test the developed model using appropriate data.
9. To compare predicted stock prices with actual values.
10. To evaluate the performance of the prediction model.
11. To improve programming and data-analysis skills.
12. To develop practical knowledge of Machine Learning and predictive modelling.
13. To understand limitations associated with stock price prediction.
14. To establish a foundation for future work in financial forecasting and Data Science.

4.5 SCOPE OF THE PROJECT

The scope covers the collection, preparation, analysis, and modelling of historical stock market data.

1. Historical Stock Data: Study opening, high, low, closing, adjusted closing prices, and volume where available.
2. Data Collection: Obtain suitable historical data from an appropriate source.
3. Data Preprocessing: Handle missing values, duplicates, formats, and data-quality issues.
4. Data Analysis: Use statistics and visualizations to understand trends and fluctuations.

5. Feature Engineering: Select or derive relevant inputs for the prediction model.
6. Machine Learning: Develop and train a predictive model.
7. Model Testing: Evaluate the model on historical observations not used for training.
8. Performance Evaluation: Compare predicted and actual values using appropriate metrics.
9. Visualization: Present historical and predicted prices through graphs.
10. Future Expansion: Extend the system with larger datasets, additional indicators, advanced time-series models, and deep learning techniques.

4.6 TECHNOLOGIES AND TOOLS USED

1. Python – primary programming language for data analysis and Machine Learning.
2. Jupyter Notebook – interactive environment for code, analysis, and visualization.
3. NumPy – numerical computation and array operations.
4. Pandas – data loading, cleaning, transformation, and analysis.
5. Matplotlib – charts and visualization.
6. Scikit-learn – preprocessing, model training, prediction, and evaluation.
7. Random Forest Regression – a suitable baseline regression model for the supplied implementation.
8. Git and GitHub – version control and project organization.
9. IDE / Code Editor – implementation and debugging.
10. Documentation Tools – preparation of project documentation.

4.7 SYSTEM ARCHITECTURE

The system architecture represents the flow of historical stock data through the prediction pipeline.

Historical Stock Data → Data Collection → Data Preprocessing → Exploratory Data Analysis → Feature Engineering → Chronological Train/Test Split → Machine Learning Model → Training → Prediction → Evaluation → Final Output

Input Data: Historical stock market data.

Preprocessing: Cleaning and preparing the dataset.

Feature Engineering: Creating lagged closing-price and rolling-average features.

Model Training: Learning relationships from the historical training portion.

Prediction: Generating the next closing-price estimate.

Evaluation: Comparing predictions with actual values using MAE, RMSE, and R^2 .

Final Output: Prediction results and an Actual-versus-Predicted chart.

4.8 METHODOLOGY

1. Problem Identification: Define the objective of predicting stock prices from historical observations.
2. Data Collection: Obtain a CSV dataset containing Date and Close, with Volume where available.
3. Data Understanding: Examine columns, data types, missing values, and chronological order.
4. Data Preprocessing: Convert dates and numeric values, remove invalid records, sort by date, and remove duplicate dates.
5. Feature Engineering: Create lagged closing prices (1, 2, 3 and 5 trading days), 5-day and 10-day moving averages, and one-day percentage return.
6. Chronological Split: Use the earlier portion for training and the later portion for testing. This avoids randomly mixing future observations into training data.
7. Model Selection: Use Random Forest Regression as a baseline model for the implementation.

8. Model Training: Train the model using historical training observations.
9. Prediction: Generate the next-day closing-price prediction for each test observation.
10. Performance Evaluation: Calculate Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.
11. Visualization: Plot actual versus predicted values for the test period.
12. Result Analysis: Review errors, trends, and limitations and identify possible improvements.
13. Final Output: Save prediction results to prediction_results.csv and the comparison plot to actual_vs_predicted.png.

4.9 EXPECTED OUTCOMES

1. Better understanding of historical stock market data.
2. Improved knowledge of data preprocessing.
3. Practical experience in analysing financial datasets.
4. Understanding of Machine Learning-based prediction.
5. Development of a stock price prediction model.
6. Improved Python programming skills.
7. Improved data visualization skills.
8. Understanding of model training and testing.
9. Knowledge of prediction performance evaluation.
10. Improved analytical and problem-solving abilities.
11. Understanding of time-dependent financial data.
12. Improved documentation and project-management skills.
13. Understanding of the limitations of historical-data-based predictions.

14. A foundation for future projects involving financial forecasting and Data Science.

Evaluation results are intentionally not fabricated in this report. They should be inserted after the supplied implementation is run on the actual historical dataset used for the project.

4.10 CERTIFICATES OF COMPLETION AND OTHER COMMUNICATION MAIL PROOF MUST BE ATTACHED

The following documents should be attached after this section:

Letter of Recommendation.



5. BIBLIOGRAPHY / REFERENCES

1. Géron, A. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow. O'Reilly Media.
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8. Scikit-learn Developers. Scikit-learn Documentation.
9. Matplotlib Development Team. Matplotlib Documentation.
10. Codec Technologies Pvt. Ltd. Internship Offer Letter, issued 20 July 2026.
11. Codec Technologies Pvt. Ltd. Internship Completion Certificate, for the internship conducted from 20 July 2026 to 20 August 2026.
12. Codec Technologies Pvt. Ltd. Letter of Recommendation, issued 20 August 2026.

APPENDIX A – PROJECT IMPLEMENTATION

A runnable Python implementation has been prepared separately. It expects a CSV file containing at minimum Date and Close columns, with Volume recommended where available.

The implementation performs preprocessing, creates lag and moving-average features, performs a chronological train/test split, trains a Random Forest Regression model, calculates MAE/RMSE/ R^2 , saves prediction_results.csv, and generates actual_vs_predicted.png.

Important: The model results must be generated from the actual historical dataset used in the student's project. No fabricated accuracy, error, or prediction figures are included.